

Yiming Lu

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EDUCATION	Emory University , Atlanta, GA Ph.D., Computer Science. GPA: 4.0/4.0 Tsinghua University , Beijing, China B.E., Automation. GPA: 3.5/4.0	August 2023 – Present Advisors: Wei Jin; Fei Liu (2023 – 2026) August 2019 – July 2023
RESEARCH INTERESTS	Self-evolving LLM agents that improve from experience through memory, skills, and execution traces; multi-agent collaboration and communication; and LLM reasoning and decision-making in high-stakes domains such as finance and public health.	
WORK EXPERIENCE	AI R&D Engineer Intern <i>Nokia</i> - Develop self-evolving agents and skills on Nokia’s internal SkillMesh platform and Merlin, an assistant that equips telecom engineers with product and domain knowledge to ramp up on or switch between projects. - Use internal agent traces to drive skill evolution, refining existing skills and distilling new ones from past executions. Intern, Internal Alpha Capture (IAC) <i>Point72</i> - Built LLM pipelines over internal analyst text data for alpha capture. - Extracted company KPIs from unstructured analyst text with LLMs and turned them into structured data. - Generated trading signals from the extracted KPIs and LLM-derived views. Research Intern, GenAI <i>Zoom Video Communications</i> - Researched cost-aware communication in LLM multi-agent teams: when agents should work solo, message asynchronously, or meet synchronously to finish shared tasks. - Built a discrete-event simulator of team workflows, evaluated on 15 software engineering workflows with 5–17 agents and three frontier LLMs; led to the first-author C2C preprint.	September 2026 – December 2026 <i>Sunnyvale, CA</i> June 2026 – August 2026 <i>New York City, NY</i> June 2025 – August 2025 <i>Bellevue, WA</i>
PUBLICATIONS	Agent Policy-Value Audit: Separating Transition Composition from Event Selection in Financial LLM Agents Mingyang Chen, Yida Xu, Huiwen Chen, Yiming Lu, Wei Jin <i>ICAIF 2026</i> [arXiv] We propose an audit that separates a financial LLM agent’s deployment value into the payoff of its action-change composition and its event-selection value, showing that an agent’s apparent gain on 723 earnings events comes from composition rather than selection. EpiEvolve: Self-Evolving Agents for Streaming Pandemic Forecasting under Regime Shifts Yiming Lu, Sihang Zeng, Zhengxu Tang, Max Lau, Fei Liu, Wei Jin <i>Under Review</i> [arXiv] We introduce EpiEvolve, a self-evolving LLM agent for streaming pandemic forecasting under regime shifts. It adapts a frozen forecaster through hierarchical episodic memory, regime-conditioned retrieval, and outcome-informed lesson distillation, without updating parameters. On weekly COVID-19 hospitalization forecasting, EpiEvolve reaches 0.629 accuracy (vs. 0.561 static; 0.325 CDC ensemble) and cuts post-shift recovery from 5 to 2 weeks. Large Language Models at Population Scale: A Survey and Taxonomy of Public Health Applications Zhengxu Tang*, Bohan Wang*, Yiming Lu*, Zewen Liu, Max S. Y. Lau, Wei Jin <i>Under Review</i> [Preprint] We present the first comprehensive survey of LLMs in public health, organized as a two-dimensional taxonomy of six public health tasks against five functional LLM roles. The analysis reveals a critical gap: current LLMs excel at extracting health signals from text but struggle with the reasoning	

required for population-level challenges, and must evolve from individual-level clinical assistants into societal-scale reasoning engines for disease dynamics, policy intervention, and health equity.

- **Communication to Completion: Modeling Collaborative Workflows with Intelligent Multi-Agent Communication**

Yiming Lu, Xun Wang, Simin Ma, Shujian Liu, Sathish Reddy Indurthi, Song Wang, Haoyun Deng, Fei Liu, Kaiqiang Song *Preprint [arXiv]*

We introduce Communication to Completion (C2C), a scalable multi-agent framework that enhances task-oriented collaboration through structured communication. C2C features the Alignment Factor for quantifying task understanding and a Sequential Action Framework for cost-aware communication decisions. On realistic coding workflows, C2C reduces task completion time by 40%.

- **STRUX: An LLM for Decision-Making with Structured Explanations**

Yiming Lu, Yebowen Hu, Hassan Foroosh, Wei Jin, Fei Liu *NAACL 2025 [Paper]*

We introduce STRUX, a framework that enhances LLM decision-making by providing structured explanations. These include favorable and adverse facts related to the decision, along with their respective strengths. On the challenging task of forecasting stock investment decisions from earnings call transcripts, STRUX outperforms strong baselines and enhances decision transparency.

- **DeFine: Enhancing LLM Decision-Making with Factor Profiles and Analogical Reasoning**

Yebowen Hu, Xiaoyang Wang, Wenlin Yao, **Yiming Lu**, Daoan Zhang, Hassan Foroosh, Dong Yu, Fei Liu *ACL 2025 Findings [Paper]*

We introduce DeFine, which builds probabilistic factor profiles from complex scenarios such as earnings calls and combines them with analogical reasoning over past experiences to guide LLM decisions under uncertainty.

PRESENTATIONS **Poster**, NAACL 2025: STRUX: An LLM for Decision-Making with Structured Explanations

TEACHING **Teaching Assistant**, Emory University
CS 571 Natural Language Processing; CS 534 Machine Learning

SERVICE Reviewer: KDD 2026 Workshop

TECHNICAL **Programming:** Python, C/C++
SKILLS **LLM Training:** PyTorch, Transformers, TRL, LLaMA Factory, VeRL, OpenRLHF